

Design and Implementation of a Graph-Orchestrated Multi-Agent System for Liquidity-Augmented, Regime-Aware Multi-Factor Analysis of Nigerian Exchange Equities

Samuel Justin Ifiezibe (sammyjayisthename@gmail.com)

Abstract

Equity factor models are widely used to explain systematic differences in stock returns, but direct transfer of conventional factor definitions to the Nigerian Exchange (NGX) is complicated by uneven trading activity, thin securities, limited point-in-time fundamental data, and changing market conditions. This paper presents the design and implementation framework of a graph-orchestrated research system for constructing and validating an NGX-specific five-factor model comprising Market, Size, Value, Momentum, and Liquidity. The system combines deterministic quantitative computation with graph-based orchestration: Python modules calculate factor returns, Newey-West statistics, bootstrap confidence intervals, Hidden Markov Model (HMM) regimes, portfolio weights, and backtest metrics, while LangGraph coordinates task execution, state transfer, and failure isolation. Financial statements are aligned to their publication dates, with an explicitly recorded fixed reporting lag used only when historical publication dates cannot be recovered, thereby limiting look-ahead bias. A three-state Gaussian HMM based on monthly market returns and rolling volatility is used to test whether factor performance changes across latent NGX regimes; weak state separation is reported as an empirical result rather than corrected by arbitrary re-specification. The applied portfolio layer standardises Size, Value, Momentum, and Liquidity signals, ranks eligible securities, forms a monthly rebalanced equal-weight long-only portfolio, applies transaction costs, and compares results with the NGX All-Share Index, CAPM, and an NGX-style Fama-French three-factor benchmark. The web implementation uses Next.js on Vercel, FastAPI on Render, and Neon PostgreSQL. The proposed platform provides a reproducible bridge between Nigerian asset-pricing research and software-engineered quantitative experimentation without allowing language models to fabricate financial evidence.

Keywords: Nigerian Exchange; multi-factor model; liquidity; Hidden Markov Model; quantitative finance; LangGraph; point-in-time data; backtesting.

1. Introduction

Multi-factor asset-pricing models explain equity returns through common market exposures and persistent firm characteristics. The Capital Asset Pricing Model (CAPM) provides the market benchmark, while the Fama-French framework demonstrates that size and book-to-market contain information not fully captured by market beta [1], [2]. Momentum adds another distinct return pattern [3], [4]. These models provide useful baselines for portfolio attribution and for determining whether apparent excess returns reflect genuine alpha or familiar systematic exposures.

The Nigerian Exchange presents a different empirical environment from the deep and continuously traded markets in which many factor models were developed. Trading activity is uneven across securities, stale prices are common in thin names, corporate fundamentals are not always available in machine-readable point-in-time form, and the number of liquid listed equities is limited. Liquidity therefore affects both the economics of expected returns and the reliability of factor measurement. Amihud's price-impact measure provides a widely used proxy for illiquidity [5], while Nigerian evidence shows that market turnover, trading activity, exchange-rate conditions, and inflation can influence equity returns [8]. Recent African frontier-market research also finds that liquidity belongs alongside conventional factors when comparing asset-pricing models across the region [7].

A second problem is computational. A factor study requires repeated data validation, point-in-time alignment, cross-sectional ranking, portfolio formation, statistical inference, regime analysis, backtesting, benchmark regressions, and provenance logging. When these steps are embedded in one long script, failures and methodological changes are difficult to isolate. The graph-engineering approach described by Roan [14] motivates a workflow in which independent factor tasks become parallel nodes, followed by validation, regime analysis, portfolio construction, and reporting. This study adapts that workflow to an academic NGX setting and removes the unsupported assumption that an AI agent should itself be responsible for numerical estimation.

The implemented research system therefore separates orchestration from computation. LangGraph manages task state, dependencies, and failure boundaries; deterministic Python functions remain the source of truth for financial calculations. The platform is designed around five factors - Market, Size, Value, Momentum, and Liquidity - and evaluates their behaviour over a configurable NGX sample, initially targeted at 2019-2025. The paper's central research question is whether a liquidity-augmented and regime-aware model provides incremental explanatory or portfolio value beyond CAPM and an NGX-style Fama-French three-factor benchmark, while the software-engineering question is whether graph orchestration makes the experiment more reproducible and auditable.

1.1 Related Work and Research Gap

Fama and French [2] established size and value as major empirical dimensions of stock returns, while Jegadeesh and Titman [3] and Carhart [4] formalised momentum as an additional return characteristic. Liquidity research shows that trading frictions can be priced and that simple turnover alone may not capture the price impact of thin trading [5]. These findings are relevant to the NGX because securities can differ sharply in turnover and trading frequency.

Evidence from Nigeria and other African frontier exchanges motivates local re-estimation rather than direct importation of developed-market results. Natufe and Evbayiro-Osagie [6] provide recent Nigerian evidence on the Fama-French three-factor model. Hearn et al. [7], using data from eight African equity markets, compare broader augmented factor models that include liquidity and momentum. Abdullahi and Fakunmoju [8] report a relationship between liquidity-related variables and Nigerian stock returns, while Ogiemudia et al. [9] show that exchange-rate and oil-price risks can materially influence Nigerian equity returns. The remaining gap is an integrated NGX research system that treats liquidity as an explicit factor, uses point-in-time fundamentals, evaluates factor stability across latent market states, and records every computational stage as a reproducible experiment.

2. Methodology and System Development

The study follows an empirical design-science approach. The asset-pricing component specifies factor construction, statistical validation, regime estimation, and portfolio rules. The software component implements those rules as modular services coordinated through a directed research graph. This design makes the numerical definitions independent of language-model prompts or orchestration choices: rerunning the same dataset and configuration must produce the same quantitative result.

2.1 High-Level System Architecture

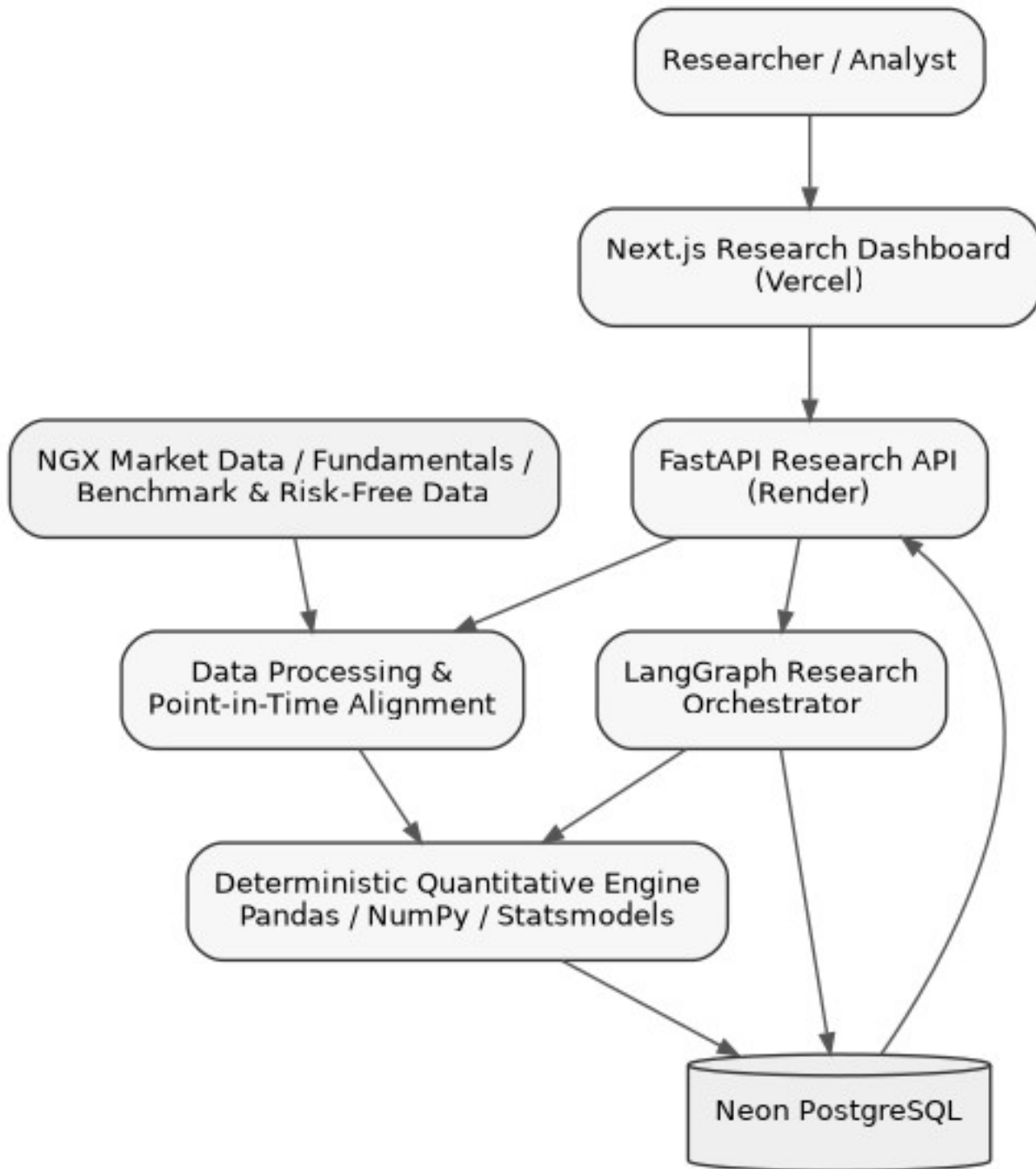


Fig 1: High-level architecture of the NGX multi-factor research system

The presentation layer is implemented with Next.js and deployed on Vercel. The application communicates with a FastAPI research API deployed on Render. The backend contains a data engine, factor modules, a statistical engine, an HMM regime engine, a portfolio and backtesting engine, and the LangGraph orchestrator. Neon PostgreSQL stores market observations, fundamentals, experiments, factor outputs, regime probabilities, and portfolio results. The graph coordinates execution, but the deterministic quantitative engine performs all factor and econometric calculations.

2.2 Data Model, Universe and Point-in-Time Alignment

The initial study targets approximately 20-30 NGX equities with adequate price history, trading activity, shares-outstanding information, and book-equity observations. The actual eligible universe is determined by data-quality rules rather than a fixed company count. Daily data provide closing prices, volume, trading value, and the NGX market benchmark.

Company fundamentals provide fiscal period, book equity, shares outstanding, and publication timing. Risk-free observations are used for excess-return calculations and CAPM regressions.

Data group	Core fields	Research use
Market prices	Ticker, date, close, volume, trading value	Returns, momentum, liquidity, backtest
Fundamentals	Fiscal period, book equity, shares outstanding, publication date	Size, value, point-in-time alignment
Market benchmark	NGX index level / return	Market factor, beta, HMM, benchmark
Risk-free data	Date, risk-free rate	Excess returns, CAPM, Sharpe calculations

Table 1: Core research data and their use

A fundamental observation is considered usable only from the date on which it became public. If a financial statement for the year ended 31 December 2023 was published on 28 March 2024, the corresponding book-equity value can influence portfolio formation only from 28 March 2024 onward. Where historical publication dates cannot be recovered, the system supports a transparent fixed reporting lag - initially 90 calendar days after fiscal year end - and records the observation as `FIXED_LAG_ESTIMATE` rather than `ACTUAL_PUBLICATION_DATE`. This rule prevents silent look-ahead bias while allowing incomplete archival data to be handled consistently.

2.3 Five-Factor Construction

The factor engine implements five independent modules. Market is used as the principal systematic benchmark; Size, Value, Momentum, and Liquidity are used both to form factor-return series and to generate cross-sectional stock characteristics. Factor modules share a common interface so their input checks and outputs can be validated independently.

Factor	Primary construction	Role in system
Market (MKT)	NGX market return minus risk-free rate	Systematic benchmark and beta control
Size (SMB)	Small-cap portfolio minus large-cap portfolio	Captures market-capitalisation effect
Value (HML)	High book-to-market minus low book-to-market	Captures relative valuation effect
Momentum (MOM)	12-1 winner portfolio minus loser portfolio	Captures medium-term return persistence
Liquidity (LIQ)	Liquidity-sorted spread using Amihud and/or turnover	NGX-specific trading-friction factor

Table 2: Implemented factor definitions

$$MKT_t = R_{m,t} - R_{f,t}$$

$$Market\ Capitalisation_{i,t} = Price_{i,t} \times Shares\ Outstanding_{i,t}$$

$$Book\text{-}to\text{-}Market_{i,t} = Book\ Equity_{i,t} / Market\ Capitalisation_{i,t}$$

$$Momentum_{i,t} = \text{cumulative return from } t-12 \text{ through } t-2$$

$$ILLIQ_i = \text{mean}(|R_{i,d}| / TradingValue_{i,d})$$

Size and Value sorts are performed using point-in-time-valid shares outstanding and book equity. Momentum excludes the most recent month to reduce short-term reversal contamination [3]. Liquidity is measured primarily with the Amihud illiquidity ratio and share turnover. Where both are available, the application may create a standardised liquidity signal in which higher turnover increases liquidity while higher Amihud price impact reduces it. The precise transformation is saved in the experiment configuration.

2.4 Statistical Validation

Each factor is passed to an independent validator that reports observation count, average return, standard deviation, annualised return and volatility, Sharpe ratio, maximum drawdown, Newey-West adjusted t-statistic, and bootstrap confidence interval. Newey and West [11] provide heteroskedasticity and autocorrelation-consistent standard errors suitable for return series that may exhibit time dependence. The study stores a working $|t|$ -statistic threshold of 2.5 as one diagnostic, but passing that threshold is not treated as sufficient evidence of economic usefulness.

Bootstrap inference uses 10,000 resamples by default. Random seeds, iteration counts, and confidence levels are persisted with the experiment so that the reported intervals can be reproduced. Statistical evidence is interpreted together with economic magnitude, turnover, drawdown, and out-of-sample behaviour. This avoids treating one favourable significance test as proof of a tradable anomaly.

2.5 Regime Detection and Diagnostics

Market regimes are estimated with a three-state Gaussian Hidden Markov Model based on monthly NGX market returns and rolling market volatility. Hidden Markov models allow the observed return process to switch between unobserved states with different statistical characteristics [10]. State labels are not defined before estimation; the fitted states are first described using average return, volatility, persistence, and transition probabilities and are then given economic labels only where the separation is meaningful.

The regime module includes explicit diagnostics because a three-state model can over-partition a small or noisy sample. The system checks the number of observations assigned to each state, convergence, state persistence, transition probabilities, and differences in state means and volatilities. If the states do not separate meaningfully, the dashboard records a weak-regime-separation finding rather than increasing the number of features or states merely to obtain a visually attractive chart. Factor returns are subsequently summarised within each valid state to determine whether apparent premia are broad or state-dependent.

2.6 Stock Scoring, Portfolio Construction and Backtesting

The applied portfolio layer converts Size, Value, Momentum, and Liquidity characteristics into cross-sectional z-scores. The baseline composite score assigns equal weight to the four ranking signals, while Market is retained as the risk and benchmark factor. Securities are ranked each month, the top N eligible stocks are selected, and the portfolio is formed using equal weights. The portfolio is long-only to reflect a practical NGX research setting in which systematic short-selling access is limited.

$$Score_{i,t} = 0.25 z(Size_{i,t}) + 0.25 z(Value_{i,t}) + 0.25 z(Momentum_{i,t}) + 0.25 z(Liquidity_{i,t})$$
$$w_{i,t} = 1 / N$$

The backtest re-estimates eligible data and signals at each monthly rebalance date, holds the selected securities for the next period, and then advances the simulation. Transaction costs are applied as a configurable basis-point cost proportional to turnover. The backtesting engine produces cumulative return, annualised return, volatility, Sharpe ratio, Sortino ratio, maximum drawdown, beta, turnover, and total simulated trading costs.

2.7 Graph-Orchestrated Execution and Implementation Stack

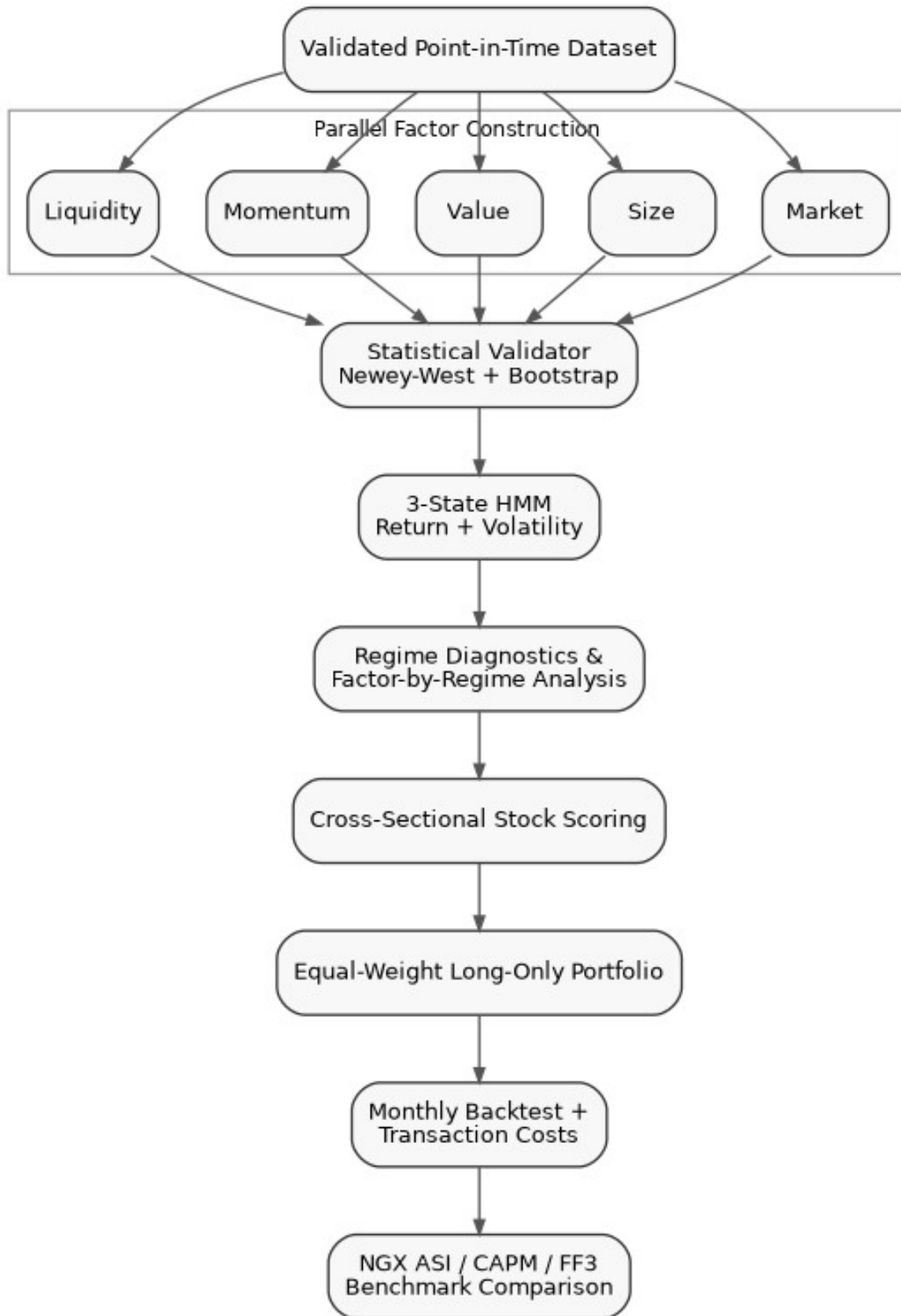


Fig 2: Graph-orchestrated quantitative research pipeline

The research graph begins after the dataset passes validation and point-in-time alignment. Five factor nodes can execute independently and then synchronise at the validator. Downstream nodes estimate market regimes, calculate regime-specific

diagnostics, score stocks, construct the portfolio, run the historical simulation, and compare the final outputs with benchmarks. Each graph node reports pending, running, completed, or failed status and writes its metadata to the experiment record. A failure in one factor calculation therefore does not erase completed work from other nodes.

The frontend is implemented in Next.js with TypeScript, Tailwind CSS, Recharts, and TanStack Query and is deployed on Vercel. FastAPI provides REST endpoints for companies, factors, regimes, portfolios, and experiments. Python 3.12 dependencies are managed with uv through pyproject.toml and uv.lock. SQLAlchemy and Alembic manage the Neon PostgreSQL schema. The numerical stack consists primarily of Pandas, NumPy, SciPy, Statsmodels, scikit-learn, and hmmlearn. LangGraph is introduced after the deterministic pipeline is functional, so orchestration cannot become a substitute for untested quantitative code.

3. Evaluation Methodology

Evaluation is divided into empirical model evaluation and software-system evaluation. The first asks whether the factor model explains or improves the analysis of NGX returns. The second asks whether the implemented platform executes that research process reliably and reproducibly. Separating the two prevents a reliable software pipeline from being mistaken for evidence of investment alpha and prevents attractive financial results from masking poor research engineering.

3.1 Empirical Evaluation

The empirical evaluation uses chronological data splits and monthly rebalancing. Where sample length permits, the research period is divided into development, validation, and untouched test windows. Factor definitions and primary modelling choices are frozen before the final test window is evaluated. The proposed model is compared with the NGX All-Share Index, CAPM, and an NGX-style Fama-French three-factor model. CAPM provides the market-only baseline; FF3 tests the incremental role of Size and Value; the proposed model then adds Momentum and Liquidity and examines factor behaviour across HMM states.

Evaluation area	Primary measures
Factor evidence	Mean return, volatility, Sharpe, Newey-West t-stat, bootstrap CI
Regime quality	State size, mean return, volatility, persistence, transition probabilities
Portfolio performance	CAGR, volatility, Sharpe, Sortino, max drawdown, turnover, net return
Model comparison	Alpha, beta exposures, R ² , benchmark-relative performance
Data integrity	Coverage, missingness, publication-date provenance, eligible-universe size
System reliability	Run reproducibility, node failures, preserved outputs, execution trace

Table 3: Evaluation dimensions and metrics

3.2 Benchmark Regressions

Portfolio excess returns are first evaluated with CAPM and then with an NGX three-factor regression containing Market, SMB, and HML. The proposed five-factor regression adds MOM and LIQ. Residual alpha is interpreted cautiously and only after transaction costs and out-of-sample results are considered. If Liquidity is statistically significant but does not reduce unexplained return or improve net portfolio performance, the study will report liquidity as explanatory rather than as a superior trading signal.

$$R_{p,t} - R_{f,t} = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{MOM} MOM_t + \beta_{LIQ} LIQ_t + \varepsilon_t$$

3.3 Software-System Evaluation

The software evaluation records whether identical data, configuration, random seed, and Git revision reproduce the same factor and backtest outputs. Each experiment stores a dataset version, factor definitions, HMM parameters, bootstrap configuration, reporting-lag policy, transaction-cost assumption, timestamps, and software commit identifier. Unit tests use manually verifiable toy datasets for Market, Size, Value, Momentum, Liquidity, portfolio weights, and selected statistical routines.

Graph orchestration is evaluated through execution traces and controlled failures. For example, a missing book-equity history can cause the Value node to fail while completed Market, Size, Momentum, and Liquidity outputs remain available. The system should identify the failed node, preserve unaffected state, and present the failure reason to the researcher. This is the practical systems benefit being tested; the graph does not create statistical evidence by itself.

4. Results and Discussion Framework

This revision does not fabricate empirical results before the NGX dataset and software implementation have been executed. The final paper will populate pre-specified factor, regime, and portfolio tables directly from persisted experiment outputs. Factor discussion will distinguish statistical significance from economic usefulness. Regime discussion will state

whether the HMM states are well separated; if they are not, weak separation will be reported as a substantive result. Portfolio discussion will emphasise net rather than gross returns and will identify whether performance is concentrated in unusually illiquid securities.

The principal comparisons are incremental. First, the study determines whether Size and Value add information beyond the market-only CAPM. Second, it determines whether Momentum and Liquidity improve explanatory power or portfolio performance beyond the NGX three-factor model. Third, it tests whether these relationships remain broadly similar across market states. Finally, any residual portfolio alpha is evaluated against transaction costs, drawdowns, turnover, and factor exposures before being described as economically meaningful.

The systems discussion will separately report data coverage, reproducibility, node-level execution status, and the degree to which graph orchestration isolates failures. A successful research run therefore has two meanings: the financial model must produce valid evidence, and the software must make the production of that evidence traceable. One cannot substitute for the other.

5. Conclusion

This paper presents the design and implementation framework of a graph-orchestrated quantitative research system for Nigerian Exchange equities. The system constructs Market, Size, Value, Momentum, and Liquidity factors from point-in-time-valid data, subjects those factors to Newey-West and bootstrap validation, estimates three latent NGX regimes using market return and volatility, ranks securities using standardised factor signals, and evaluates a monthly rebalanced equal-weight long-only portfolio against the NGX All-Share Index, CAPM, and an NGX-style three-factor benchmark.

The design deliberately treats agent orchestration as a software-engineering mechanism rather than a source of financial truth. LangGraph controls dependencies, state, and failure boundaries, while deterministic Python routines perform all numerical calculations. Next.js, FastAPI, Render, Vercel, and Neon provide the application and deployment layers needed to expose the research process as a reproducible web system. The resulting platform is intended to generate the empirical evidence required to evaluate whether liquidity and regime dependence materially improve multi-factor analysis in the Nigerian equity market.

5.1 Future Work

Future extensions can introduce profitability, investment, and low-volatility factors; additional liquidity proxies; inflation, exchange-rate, oil-price, and monetary-policy regime features; inverse-volatility or constrained optimisation; richer transaction-cost models; and automated research-report generation. These extensions are supported by the modular architecture without changing the definitions of the five factors evaluated in the core study.

5.2 Core Contribution

The core contribution is an integrated and auditable NGX research platform that combines liquidity-aware factor construction, point-in-time data controls, regime diagnostics, portfolio backtesting, and graph-based orchestration. Its methodological claim is intentionally limited: AI-assisted orchestration can improve the organisation and traceability of quantitative research, but all claims about factor significance, market regimes, and portfolio performance must remain grounded in deterministic calculations and observed data.

Declarations

Funding: No external funding has been declared.

Conflict of interest: The author declares no conflict of interest.

Data availability: The study uses NGX market and corporate data together with benchmark and risk-free series. Redistribution will depend on the licences attached to the source datasets.

Code availability: The implementation is intended for release in a version-controlled repository after removal of credentials and any non-redistributable data.

Ethics approval: Not applicable. The study uses secondary financial and corporate data and involves no human participants.

References

- [1] Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425-442.
- [2] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3-56. [https://doi.org/10.1016/0304-405X\(93\)90023-5](https://doi.org/10.1016/0304-405X(93)90023-5)
- [3] Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1), 65-91.
- [4] Carhart, M. M. (1997). On persistence in mutual fund performance. *The Journal of Finance*, 52(1), 57-82.
- [5] Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31-56. [https://doi.org/10.1016/S1386-4181\(01\)00024-6](https://doi.org/10.1016/S1386-4181(01)00024-6)
- [6] Natufe, O. K., & Evbayiro-Osagie, E. I. (2024). Fama and French three-factor model: Evidence from the Nigerian Stock Exchange. *Finance and Banking Review*, 19(1), 1-23.
- [7] Hearn, B., Taurigana, V., Ntim, C. G., Malagila, J. K., & Mishra, T. (2025). Asset pricing in African frontier equity markets. *International Review of Financial Analysis*, 97, 103752. <https://doi.org/10.1016/j.irfa.2024.103752>

- [8] Abdullahi, I. B., & Fakunmoju, S. K. (2019). Market liquidity and stock return in the Nigerian Stock Exchange market. *Binus Business Review*, 10(2), 87-94. <https://doi.org/10.21512/bbr.v10i2.5588>
- [9] Ogiemudia, O. A., Osifo, O., & Eghosa, I. L. (2022). Market risk factors and stock returns in the Nigerian bourse. *CBN Journal of Applied Statistics*, 13(2), 79-115. <https://doi.org/10.33429/2476-8472.1252>
- [10] Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357-384.
- [11] Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703-708.
- [12] Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1-26.
- [13] Bailey, D. H., & López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5), 94-107.
- [14] Roan [@RohOnChain]. (2026, July 23). How to use graph engineering to build a multi-factor alpha model [X article]. <https://x.com/i/status/2080296261576687751>
- [15] Wu, Q., Bansal, G., Zhang, J., et al. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. arXiv:2308.08155.
- [16] Yao, S., Zhao, J., Yu, D., et al. (2023). ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations (ICLR)*.
- [17] Nigerian Exchange Group. (n.d.). Equities price list and market data. <https://ngxgroup.com/exchange/data/equities-price-list/>
- [18] Central Bank of Nigeria. (n.d.). Rates and financial statistics. <https://www.cbn.gov.ng/rates/>